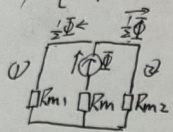


(一) 磁路
1-7 [板-圆孔, 按部就班] 板]



1. 中间铁心柱磁通为 $7.5 \times 10^{-4} \text{ Wb}$, 不计铁心的磁位降时, 所需的直流励磁电流。

$$F = \sum H l_i \quad H = \frac{B}{\mu} \quad \mu = 4\pi \times 10^{-7} \text{ H/m}, \text{ 求 } B.$$

$$\oint \vec{B} \cdot d\vec{l} = \Phi.$$

支路①: $\frac{1}{2}\Phi$, $A_1 = \frac{(7.5 - 1.4 \times 2 - 2.5)}{2} + 8) (0.93 \times 2.5 + 8) = 1.215 \times 2.35 \text{ cm}^2 = 2.99625 \text{ cm}^2$

支路②: 同①. $B_2 = \frac{\frac{1}{2}\Phi}{A_2} = \frac{\frac{1}{2}\Phi}{A_1}$

干路: Φ , $A = (2.5 + 8) \times (0.93 \times 2.5 + 8) = 5.934 \text{ cm}^2 \approx 6 \text{ cm}^2 = 6 \times 10^{-4} \text{ m}^2$

$$F = H_1 l_1 + H_2 l_2 + H_l$$

$$= \frac{B_1}{\mu} \cdot l_1 + \frac{B_2}{\mu} \cdot l_2 + H_l$$

$$= \frac{\frac{1}{2}\Phi}{A_1 \cdot \mu} l_1 + \frac{\frac{1}{2}\Phi}{A_2 \cdot \mu} l_2 + \frac{\Phi}{A \mu} l$$

$$= \frac{\Phi}{2A_1 \cdot \mu} l_1 + \frac{\Phi}{A \mu} l$$

$$2A_1 = A, l_1 = l = 0.025 \text{ cm}$$

$$\Rightarrow \frac{2\Phi}{A\mu} \times l = \frac{2 \times 7.5 \times 10^{-4}}{6 \times 10^{-4} \times 4\pi \times 10^{-7}} \times 0.025 \times 10^{-2} = 2.08 \text{ A}$$

$$F = N i$$

$$N = 500, i = \frac{497}{500} \approx 1 \text{ A}$$

例1-3 P15

(二) 直流电机的运行

1. 他励直流电动机的 $R = 100 \text{ k}\Omega$, 效率为 70% , $U_N = 500 \text{ V}$, $I_N = 10 \text{ A}$, $n_N = 1000 \text{ r/min}$, 已知电枢电阻为 0.05Ω , 求: (1) 电动机的电枢电流 (2) 电动机的反电势 (3) 电动机的输出转矩。

$$\eta = \frac{P_2}{P_1} = \frac{U_N I_a}{U_N I_a}, I_a = 222.2 \text{ A}$$

$$\text{电动机 } E_a = C_e \Phi n = U_N - I_a R_a$$

$$= 500 \text{ V} - 222.2 \times 0.05 \Omega$$

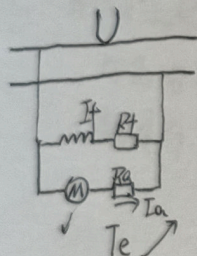
$$= 488.89 \text{ V}$$

3. 电动机的输出转矩。

$$T_2 = T_{\text{电磁}} \quad \Omega_N = \frac{n_N \times 2\pi}{60} = 104.67 \text{ rad/s}$$

$$= \frac{P_2}{\Omega_N} = \frac{P_N}{\Omega_N} = \frac{P_N}{104.67 \text{ rad/s}} = \frac{100 \text{ kW}}{104.67 \text{ rad/s}} = 955.38 \text{ N}\cdot\text{m}$$

分析题: 一台并励直流电机, 如果电源电压和励磁电流保持不变, 加上恒定转矩的负载后, 发现电枢电流远大于额定电流, 有人试图在电枢回路中串联电阻来限制电枢电流, 问是否可行? 为什么?



$I_a = I_N$, 串联电阻。

I_f 始终不变, $\Phi = k I_f$ 不变。

起初, n 未变化, $E_a = U - I_a'(R_a + R_2)$

$$E_a = C_e \Phi n, \text{ 不变}$$

I_a' 减小, 达到减小电流目的。

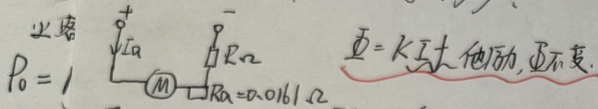
但稳定后 $T_e = C_T \Phi I_a$

Φ 不变, T_e 不变, 则 I_a' 必须大于原 I_a , 则不可行

$$k_{st} = \frac{T_{st}}{I_{st}} = \frac{T_{max}}{I_N}$$

(二) 直流电机

例2-1, 一台他励电动机的额定数据为: $P_N = 13.5 \text{ kW}$, $U_N = 750 \text{ V}$, $I_N = 19.30 \text{ A}$, $n_N = 2000 \text{ r/min}$.
电枢绕组的电阻 $R_a = 0.0161 \Omega$, 电刷接触电压降 $2\Delta V_c = 2 \text{ V}$. 设电动机在较速 2000 r/min 和额定负载下运行, 在负载的总制动转矩 (包括损耗转矩) 保持不变的条件下, 试求:
① 在电枢回路中突然串入电阻 $R_2 = 0.0743 \Omega$, 在串入电阻的最初瞬间和达到稳态时的电枢电流和转速分别为多少? (转速尚未必要以同下册的分析为准)



最初: n 不变, $E_a = C_e \Phi n$, 不变
 $E_a = U_N - I_a R_a = 750 \text{ V} - 19.30 \text{ A} \times 0.0161 \Omega$
 $I_a (R_a + R_2) = U_N - E_a = 718.9 \text{ V}$
 最初 $I_a = \frac{U_N - E_a}{R_a + R_2} = \frac{750 \text{ V} - 718.9 \text{ V}}{0.0161 \Omega + 0.0743 \Omega} = 344 \text{ A}$

稳态: 负载 $T_e = C_T \Phi I_a$, 要保持不变, I_a 要回到原来, 稳态时电枢转速回到原来的数值, 电枢电流要回到原来
 $I_a = 19.30 \text{ A}$, $E_a = U_N - I_a (R_a + R_2) = 575.528 \text{ V}$
 $E_a = C_e \Phi n_{\text{稳}}$, $\frac{n_{\text{稳}}}{n} = \frac{E_{a\text{稳}}}{E_a} = \frac{575.528 \text{ V}}{718.9 \text{ V}}$, $n_{\text{稳}} = 1601 \text{ r/min}$

② 减少电动机的励磁电流, 使磁通减少 10%, 当达到稳态时的电枢电流和转速各为多少?

$\Phi = k \cdot I_f$, $\Phi' = 0.9 \Phi$

$E_a = C_e \Phi n$
 $T_e = C_T \Phi I_a$, $I_a' = \frac{10}{9} I_a$, T_e 保持不变
 $= 2444.4 \text{ A}$

又 $E_a' = U_N - I_a' \cdot R_a$

$C_e \Phi' n' = 750 \text{ V} - 2444.4 \text{ A} \times 0.0161 \Omega$
 $= 715.47 \text{ V}$

$\frac{E_a'}{E_a} = \frac{715.47 \text{ V}}{718.9 \text{ V}} = \frac{\Phi' \cdot n'}{\Phi \cdot n} = 0.9 \times \frac{n'}{n}$
 变化不大
 $n' = 221 \text{ r/min}$

(二) 直流电机 功率图

2-15 一台并励直流电机在额定电压 $U_N = 220V$ 和额定电流 $I_N = 80A$ 的情况下运行, 电枢回路总电阻 $R_a = 0.08\Omega$, $2\Delta U_c = 2V$, 励磁回路总电阻 $R_f = 88.8\Omega$, 额定负载时的效率 $\eta_N = 85\%$. 试求:

1) 额定输入功率

$$P_1 = U_N \cdot I_N$$

2) 额定输出功率

$$P_0 = \eta_N \cdot P_1 = P_N = 85\% \times P_1$$

3) 总损耗

$$(1 - \eta_N) \times P_1 = P_{\text{total}}$$

$$P_{\text{arm}} = I_f^2 R_f = \frac{U_N^2}{R_f}$$

$$I_f = \frac{U_N}{R_f} = \frac{220V}{88.8\Omega} =$$

$$I_a = I_N - I_f$$

$$P_{\text{arm}} = I_f^2 \cdot R_f$$

$$P_{\text{arm}} = I_a^2 \cdot R_a$$

$$1) \text{ 电刷接触损耗 } P_A = I_a \cdot 2\Delta U_c$$

$$P_1 - P_{\text{arm}} = P_{\text{mech}}$$

$$1) \text{ 附加损耗}$$

$$P_{\text{total}} = P_A = 1\% P_N$$

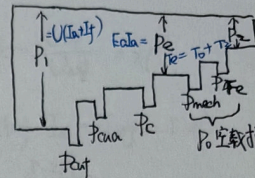
$$2) \quad P_{\text{mech}} + P_{\text{Fe}} = P_1 - P_{\text{arm}} - P_A - P_{\text{Cu}} = P_1 - P_{\text{arm}} - P_A - P_{\text{Cu}}$$

$$P_{\text{Fe}} = P_2$$

$$= P_e - P_N$$

$$P_e = E_a I_a - P_N$$

$$E_a = U_N - I_a R_a$$



直流电机励磁

电流恒定

铁芯磁阻铁耗

而于铁芯磁阻铁耗

但并励电机中,

转子受电源影响,

为短路,

过一次励磁电流, 存在铁耗

2-10 并励直流电动机

并励直流电动机的起动电流决定于什么? 正常工作时电枢电流又决定于什么?

电压与电枢电流, 在额定电压下, 直流电动机的起动电流 I_a 决定于电枢回路的总电阻。

转速与电压

起动电流会较大, 随着转速的上升和感应电动势的增加, 电枢电流会迅速下降。

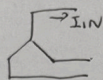
正常工作时, 直流电动机的电枢电流的大小取决于负载。

$$T_e = C_T \Phi I_a$$

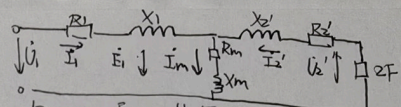
(电机的机械特性与负载特性的交点)

(三) 变压器

一台Y系列的三相电力变压器，高、低压均为Y接， $S_N = 200 \text{ kVA}$ ， $\frac{U_{1N}}{U_{2N}} = \frac{10}{0.4} \text{ kV}$ ， $\frac{I_{1N}}{I_{2N}} = \frac{11.5}{288.7} \text{ A}$ 。
在低压侧施加额定电压做空载试验，测得 $P_0 = 470 \text{ W}$ ， $I_0 = 0.018 \times I_{2N} = 5.2 \text{ A}$ ，求励磁参数。
计并高、低压侧相电压。



T形等效



$$\begin{aligned} \dot{E}_2' &= \dot{E}_1 \\ \dot{E}_2 &= \dot{E}_1 \\ &= I_2' Z_2' + \dot{U}_2' \end{aligned}$$

$$U_{1\phi} = \frac{S_N}{3 \times I_{1N} / \sqrt{3}} = 5773.5 \text{ V}$$

$$\text{1为高压侧, 2为低压侧} \quad k = \frac{U_{1\phi}}{U_{2\phi}} = 25$$

$$U_{2\phi} = \frac{S_N}{3 \times I_{2N} / \sqrt{3}} = 230.9 \text{ V}$$

$$I_0 = 5.2 \text{ A 为高压侧(低压侧)} \quad I_0 = I_{2\phi} = 5.2 \text{ A}$$

$$I_{1\phi} = I_{2\phi} \times \frac{1}{k} = \frac{5.2 \text{ A}}{25}$$

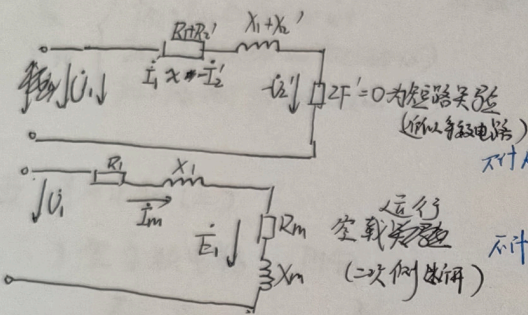
$$\text{每相铁耗 } P_{0\phi} = \frac{470 \text{ W}}{3} = 156.7 \text{ W}$$

$$\text{铁耗 } P_{0\phi} = P_{Fe} = 156.7 \text{ W} = I_0^2 \times R_m$$

$$R_m = \frac{U_{2\phi}}{I_{20\phi}} = \frac{230.9}{5.2} \Omega = 44.4 \Omega$$

$$X_m = \sqrt{Z_m^2 - R_m^2} = 44 \Omega$$

$$\frac{k^2}{R_m} \rightarrow \frac{Z_m}{X_m}$$



(空载电路)

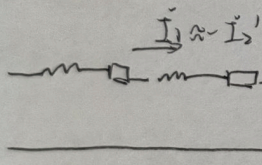
不计 R_2 铁耗

空载试验

不计 R_2 铜耗

(二次侧开路)

分析题：变压器稳定运行时，一次侧电压及其频率均为额定值，其他条件不变，当负载阻值变大时(电阻负载)，一次侧电流将如何变化，二次侧电流将如何变化，主磁通将如何变化，并解释原因。



高

电流 $I_1 \downarrow$

电流 $I_2 \downarrow$ (由产生原因)

主磁通几乎不变，

因为 I_m 几乎不变，则 $\dot{E}_1 \approx \dot{U}_1$ ，而 $\dot{U}_1$ 不变

R_m, X_m

$\downarrow$
 I_m 不变

四. 异步机 (-)

1. 同步转速 $n_s = \frac{60f}{p}$

转差率 $s = \frac{n_s - n}{n_s}$

2. 定子旋转磁场的形成

P17 工作原理介绍

(4-30) (4-31) (4-32) 旋转磁场方向、幅值

$i_{A1} = I_{\phi 1} \cos \frac{\pi}{6} \cos \omega t$

$i_{B1} = I_{\phi 1} \cos (\frac{\pi}{6} - 120^\circ) \cos (\omega t - 120^\circ)$

$i_{C1} = I_{\phi 1} \cos (\frac{\pi}{6} + 120^\circ) \cos (\omega t - 240^\circ)$

$F_1 = \frac{3}{2} I_{\phi 1}$

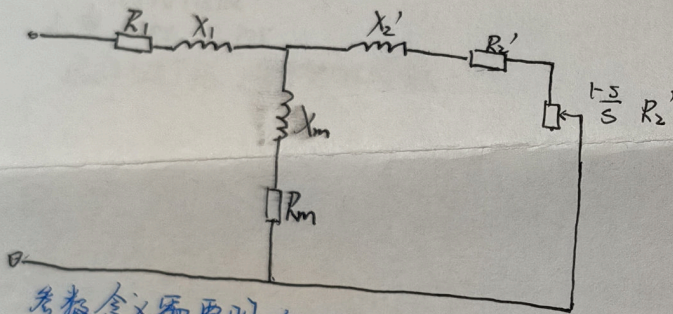
$i_1(x, t) = i_{A1} + i_{B1} + i_{C1}$

4-36.

$= \frac{3}{2} I_{\phi 1} \cos (\omega t - \frac{\pi}{6})$

五. 异步电机 (=)

1. T型等效电路, P142

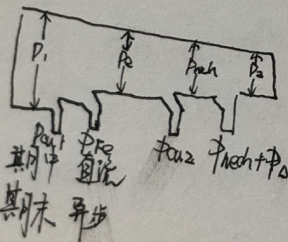


参数含义需要明确知道

2. 功率、转矩. P146

功率、转矩.

功率流程图



计算功率

计算转矩、转速

$P_i = P_{cu1} + P_{Fe} + P_e$

$P_e = P_{cu2} + P_{mech}$

$P_{mech} = P_0 + P_{mech} + P_2$

$P_{cu2} = s P_e$

$P_{mech} = (1-s) P_e \rightarrow P_e = P_e$

四. 异步电机 (-)

1. 验证 2. 分析 3. 计算 4. 画图

4-16. 若存在对称两相绕组 (两绕组的匝数相同在空间相差 90° 电角度) 中通入对称两相电流 $i_A = I_m \cos \omega t$, $i_B = I_m \sin \omega t$.

用数学分析法证明两相合成磁动势为旋转磁动势 具体数值 (三相)

$$i_A = I_m \cos \omega t = I_m \sin(\omega t - 90^\circ)$$

$$i_B = I_m \sin \omega t$$

$$i_A = I_m \cos(\frac{\pi x}{\tau}) \sin(\omega t - 90^\circ)$$

$$i_B = I_m \sin(\frac{\pi x}{\tau}) \sin \omega t$$

$$f_{A1} = F_1 \cos(\frac{\pi x}{\tau}) \cos \omega t$$

$$f_{B1} = F_1 \sin \pi$$

$$f_{B1} = F_1 \cos(\frac{\pi x}{\tau} - 90^\circ) \cos(\omega t - 90^\circ)$$

$$= F_1 \sin(\frac{\pi x}{\tau}) \sin(\omega t)$$

$$f_{A1} + f_{B1} = F_1 \cos(\frac{\pi x}{\tau}) \cos(\omega t) + F_1 \sin(\frac{\pi x}{\tau}) \sin(\omega t)$$

$$= F_1 [\cos(\frac{\pi x}{\tau}) \cos(\omega t) + \sin(\frac{\pi x}{\tau}) \sin(\omega t)]$$

$$= F_1 \cos(\frac{\pi x}{\tau} - \omega t)$$

$$= F_1 \cos(\omega t - \frac{\pi x}{\tau})$$

两相合成磁动势为圆形旋转磁动势

绕组集中分布, 由整距及为短距的长处?

1. 集中 $\rightarrow$ 分布 $11/4$

削弱谐波

2. 整 $\rightarrow$ 短 15

选择合适节距, 削弱特定次谐波

课本. 三相磁动势推导,

取 A 相绕组的轴线处作为空间坐标的原点, 并以正相序方向作为 x 的正方向,

同时选择 A 相电流达到最大值的瞬间作为时间的起始点, 则 A、B、C 三相的基波磁动势表示为

$$f_{A1} = F\phi_1 \cos \frac{\pi x}{\tau} \cos \omega t$$

$$f_{B1} = F\phi_1 \cos(\frac{\pi x}{\tau} - 120^\circ) \cos(\omega t - 120^\circ)$$

$$f_{C1} = F\phi_1 \cos(\frac{\pi x}{\tau} - 240^\circ) \cos(\omega t - 240^\circ)$$

利用公式 $\cos \alpha \cos \beta = \frac{1}{2} [\cos(\alpha - \beta) + \cos(\alpha + \beta)]$

$$f_{A1} = \frac{1}{2} F\phi_1 \cos(\frac{\pi x}{\tau} - \omega t) + \frac{1}{2} F\phi_1 \cos(\frac{\pi x}{\tau} + \omega t)$$

$$f_{B1} = \frac{1}{2} F\phi_1 \cos(\frac{\pi x}{\tau} - \omega t) + \frac{1}{2} F\phi_1 \cos(\frac{\pi x}{\tau} + \omega t - 240^\circ)$$

$$f_{C1} = \frac{1}{2} F\phi_1 \cos(\frac{\pi x}{\tau} - \omega t) + \frac{1}{2} F\phi_1 \cos(\frac{\pi x}{\tau} + \omega t - 120^\circ)$$

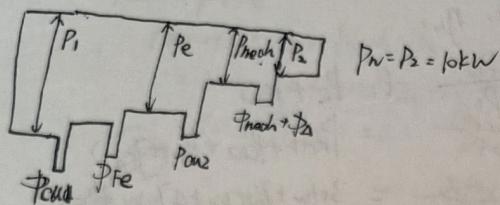
基波合成磁动势 $f_{A1} + f_{B1} + f_{C1} = \frac{3}{2} F\phi_1 \cos(\omega t - \frac{\pi x}{\tau})$

$$F_1 = \frac{a}{2} \frac{p}{p} \frac{N}{p} I$$

五 异步电动机(二)

3φ, p=2, 异步电动机, $P_N = 10 \text{ kW}$, $U_N = 380 \text{ V}$, $f_N = 50 \text{ Hz}$, 星形接法,
额定运行时, $P_{Cu} = 200 \text{ W}$, $P_{Fe} = 100 \text{ W}$, $P_{Cu2} = 100 \text{ W}$, $\Phi \Phi_{mech} + \Phi_{\Delta} = 200 \text{ W}$ 。

功率流程图



$$\begin{aligned} P_1 &= m U_1 I_1 \cos \varphi_1 \\ P_{Cu1} &= m I_1^2 R_1 \quad P_{Fe} = m I_m^2 R_m \\ P_e &= m I_2' I_2' \cos \varphi_2' = m I_2'^2 \frac{R_2'}{s} \\ P_{Cu2} &= m I_2'^2 R_2' \\ P_{Cu2} &= s P_e \\ P_{mech} &= P_e - P_{Cu2} = m I_2'^2 \frac{R_2'}{s} - m I_2'^2 R_2' \\ &= (1-s) m I_2'^2 \frac{R_2'}{s} \\ &= (1-s) P_e \end{aligned}$$

同步转速 n_s

$$n_s = \frac{60 f_N}{p} = \frac{3000}{2} = 1500$$

$$T_e = T_2 + T_0, \quad T_0 = \frac{P_{mech} + P_{\Delta}}{\omega}$$

$$\frac{P_e}{\omega_s} = \frac{P_{mech}}{\omega} = T_e$$

$$\therefore P_e = \frac{\omega}{1-s} (P_{mech} + P_{\Delta}) + P_{Cu2}$$

$$= \frac{1}{1-s} (10000 + 200) + 100 = 10300 \text{ W}$$

4) s, n_N

$$s P_e = P_{Cu2}, \quad s = \frac{P_{Cu2}}{P_e} = 0.0097$$

$$\begin{aligned} n_N &= (1-s) n_s \\ &= 1485.5 \text{ rpm} \end{aligned}$$

5) T_e

$$T_e = \frac{P_e}{\omega_s}$$

$$\begin{aligned} \frac{P_e}{\omega_s} &= \frac{10300}{1500 \times \frac{2\pi}{60}} = 155.48 \text{ rad/s} \\ T_e &= 66.246 \text{ N}\cdot\text{m} \end{aligned}$$

$$16) \eta_N = \frac{P_2}{P_1} = \frac{P_N}{P_e + P_{Cu1} + P_{Fe}} = \frac{10000 \text{ W}}{10300 \text{ W} + 100 + 200} = \frac{10000 \text{ W}}{10600 \text{ W}} = 94.3\%$$

$$\begin{aligned} 17) T_2 &= \frac{P_{mech}}{\omega} = \frac{P_e}{\omega_s} \quad T_2 = \frac{P_2}{\omega} \\ \omega &= \frac{n_N \times 2\pi}{60} = \frac{1485.5}{9.55} = 155.5 \text{ rad/s} \\ T_2 &= \frac{10000 \text{ W}}{155.5 \text{ rad/s}} = 64.288 \text{ N}\cdot\text{m} \end{aligned}$$

5-11. 三相异步电动机的铭牌: $P_N = 10\text{kW}$, $U_N = 380\text{V}$, $I_N = 11\text{A}$

异步电动机 (二)

5-13 一台三相异步电动机, $U_N = 380\text{V}$, $f_N = 50\text{Hz}$, $P_N = 7.5\text{kW}$, $n_N = 962\text{r/min}$

定子绕组为三角形联结, $\cos\varphi_N = 0.827$, $P_{cu1} = 470\text{W}$, $P_{fe} = 234\text{W}$, $P_{mech} = 95\text{W}$, $P_s = 80\text{W}$,
试求额定运行时的:

1. 转差率 s ;

$$n_s = 1000 \text{ r/min} = \frac{60f}{p} = \frac{60 \times 50}{3} = 1000 \text{ r/min}$$

$$s = \frac{n_s - n_N}{n_s} = \frac{38}{1000} = 0.038$$

2. 额定电流 I_N

$$f_2 = s f_1 = 0.038 \times 50\text{Hz} = 1.9\text{Hz}$$

3. 铜耗

$$P_{cu} = s P_e = 0.038 \times P_e = 0.038 \times \frac{P_{mech}}{1-s}$$

$$P_e = \frac{P_N}{1-s} = \frac{7.5}{1-0.038} = 7.81\text{kW}$$

$$P_{cu} = 0.038 \times 7810 = 297\text{W}$$

4. 效率 η_N :

$$\eta_N = \frac{P_N}{P_1}$$

$$P_1 = P_e + P_{cu1} = P_{mech} + P_{cu2} + P_{cu1} + P_{fe}$$

$$= 301\text{W} + 162\text{W} + 470\text{W} + 234\text{W}$$

$$= 863\text{W}$$

$$\eta = \frac{P_N}{P_1} \times 100\% = 87\%$$

$$I_1 = \frac{P_1}{\sqrt{3} U_N \cos\varphi_N}$$

$$P_1 = 3 U_N I_1 \cos\varphi_N$$

$$= 3 \times 380\text{V} \times I_1 \times 0.827$$

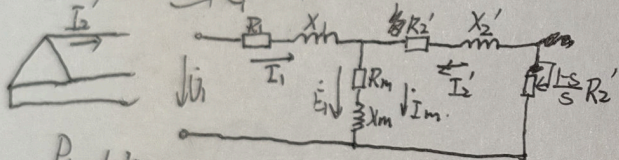
$$I_1 = \frac{P_1}{\sqrt{3} U_N \cos\varphi_N} = 11\text{A}$$

5-14. 三相异步电动机

每相参数: $P_N = 10\text{kW}$, $U_N = 380\text{V}$, $2p = 4$, $f_N = 50\text{Hz}$, $n_N = 1440\text{r/min}$

$R_1 = 1.375\Omega$, $X_1 = 2.43\Omega$, $R_2' = 1.04\Omega$, $X_2' = 4.4\Omega$, $R_m = 8.34\Omega$, $X_m = 82.6\Omega$

定子绕组为三角形联结, 在额定负载时的 $P_{mech} + P_0 = 105\text{W}$, 求额定负载时的 I_1 、功率因数、输入功率及效率。



$$P_N = 10\text{kW}$$

$$P_{mech} = P_N + P_{mech} + P_0$$

$$= 10000 + 105\text{W} = 10105\text{W}$$

$$n_s = \frac{60f_N}{p} = 1500\text{r/min}$$

$$s = \frac{n_s - n_N}{n_s} = \frac{45}{1500} = 3\%$$

$$X_2' + \frac{R_2'}{s} = 34.94 \angle 7.23^\circ$$

$$Z = R_1 + jX_1 + (X_m + R_m) \parallel (X_2' + \frac{R_2'}{s})$$

$$= 1.375\Omega + j2.43\Omega + (83.2 \angle 84.23^\circ \parallel 34.94 \angle 7.23^\circ)$$

$$= 32.27 \angle 30.44^\circ$$

定子电流:

$$I_1 = \frac{U_N}{Z} = \frac{380 \angle 0^\circ}{32.27 \angle 30.46^\circ} = 11.77 \angle -30.46^\circ \text{ 相电流}$$

功率因数: $\cos\varphi_N = \cos 30.46^\circ = 0.86$

$$|I_1| \cdot \sqrt{3} = 20.38\text{A}$$

$$P_1 = \sqrt{3} U_N I_1 \cos\varphi_1 = 11539\text{W}$$

$$\eta = \frac{P_2}{P_1} = \frac{10\text{kW}}{11539\text{W}} \times 100\% = 86.6\%$$

5-12

五、异步电动机(二)

5-15. 三相异步电动机铭牌数据: $P_N=10\text{kW}$, $U_N=380\text{V}$, $I_N=11\text{A}$.

定、转子绕组为星形联结, 额定运行时, $P_{cu1}=157\text{W}$, $P_{cu2}=314\text{W}$, $P_{Fe}=276\text{W}$, $P_{mech}=77\text{W}$, $P_d=200\text{W}$.

求: 1. 额定转速

$$n_0 = \frac{60f_1}{p} = \frac{60 \times 50}{2} = 1500 \text{ r/min}$$

求: 2. s

$$P_{cu2} = s P_{Fe}$$

$$P_{Fe} = P_{cu1} + P_{cu2} + P_{mech} + P_d$$

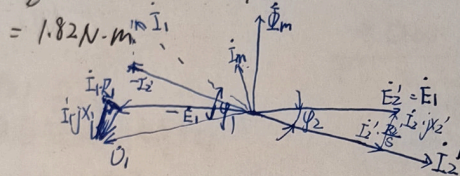
$$= 1000\text{W} + 314\text{W} + 77\text{W} + 200\text{W}$$

$$= 10591\text{W} \quad s = \frac{P_{cu2}}{P_{Fe}}$$

$$= \frac{314\text{W}}{10591\text{W}} = 0.03 = 3\%$$

$$\alpha T_0 = \frac{P_0}{\omega_n} = \frac{77\text{W} + 200\text{W}}{1500 \times \frac{2\pi}{60}}$$

$$\alpha T_0 = \frac{276\text{W}}{1500 \times \frac{2\pi}{60}} = 1.82 \text{ N}\cdot\text{m}$$



$$n = \frac{1-s}{s} n_s \quad (r/s)$$

$$= 1455 \text{ r/min}$$

3. T_e

$$T_e = \frac{P_e}{\omega_s} = \frac{P_e}{\frac{2\pi n_s}{60}}$$

$$= \frac{10591\text{W}}{1500 \times \frac{2\pi}{60}} = 67.5 \text{ N}\cdot\text{m}$$

$$\phi T_2 = T_e + T_0$$

$$= 67.5 \text{ N}\cdot\text{m} + 1.82 \text{ N}\cdot\text{m}$$

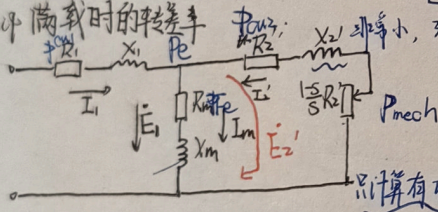
$$= 69.3 \text{ N}\cdot\text{m}$$

5-16. 三相绕线转子 $U_N=380\text{V}$, $f_N=50\text{Hz}$. 定、转子绕组为星形联结, $n_N=1440\text{r/min}$.

每相参数 $R_1=0.4\Omega$, $R_2'=0.4\Omega$, $X_1=1\Omega$, $X_2'=1\Omega$, $X_m=40\Omega$, R_m 略去不计. 设转子为三相.

定、转子绕组的有损电阻比为 4, 试求:

1. 满载时的转差率



$$s = \frac{n_0 - n_N}{n_0} = \frac{1500 - 1440}{1500} = 4\%$$

$$s = \frac{n_0 - n_N}{n_0} = \frac{1500 - 1440}{1500} = 4\%$$

2. 解由 I_1 , I_2 和 I_m

$$Z_1 = R_1 + jX_1 = 0.4 + j1 = 1.077 \angle 68.2^\circ$$

$$Z_2 = \frac{R_2'}{s} + jX_2' = \frac{0.4}{0.04} + j1 = 10.77 \angle 5.3^\circ$$

$$Z_m = R_m + jX_m = j40$$

$$Z = Z_1 + Z_2 \parallel Z_m = 1.077 \angle 68.2^\circ + 10.77 \angle 19.99^\circ$$

$$= 10.91 \angle 24.2^\circ$$

$$I_1 = \frac{U_N}{Z} = \frac{380 \angle 0^\circ}{10.91 \angle 24.2^\circ} = 34.83 \angle -24.2^\circ$$

$$\varphi_1 = 24.2^\circ, \cos \varphi_1 = 0.912$$

4. 总机械功率 $P_{mech} = 3 I_2^2 \times \left(\frac{R_2'}{s} \right)$

$$= 3 \times (19.2)^2 \times \frac{0.4}{0.04} = 1172 \text{ W}$$

$$T_e = \frac{P_e}{\omega_s} = \frac{P_{mech}}{\omega_s} = \frac{1172}{1500 \times \frac{2\pi}{60}} = 7.5 \text{ N}\cdot\text{m}$$

$$P_1 = m_1 U_1 I_1 \cos \varphi_1 = 3 \times 380 \times 34.83 \times 0.912 = 36212 \text{ W}$$

$$P_e = m_1 I_2^2 \times \left(\frac{R_2'}{s} \right) = m_1 I_2^2 \times \frac{R_2'}{s}$$

$$E_2' = I_2 \left(\frac{R_2'}{s} + jX_2' \right) = 19.2 \angle -7.2^\circ \times (10.77 \angle 5.3^\circ) = 207.6 \angle -1.9^\circ$$

$$E_2' = k_e E_2 = k_e \frac{1}{k_e} E_1 = E_1$$

$$I_2 = \frac{U_N}{R_2' + jX_2' + R_1 + jX_1} = \frac{380 \angle 0^\circ}{10.91 \angle 24.2^\circ} = 34.83 \angle -24.2^\circ$$

$$I_m = \frac{U_N}{X_m} = \frac{380}{40} = 9.5 \angle -90^\circ$$

$$I_2' = \frac{U_N}{R_2' + jX_2' + R_1 + jX_1} = \frac{380 \angle 0^\circ}{10.91 \angle 24.2^\circ} = 34.83 \angle -24.2^\circ$$

$$I_1' = I_2' + I_m = 21.18 \angle -25^\circ \text{ A}$$

$$E_{2s} = s E_2' = 0.04 \times 207.6 = 8.3 \text{ V}$$

$$E_2' = k_e E_2 = k_e \frac{1}{k_e} E_1 = E_1$$

$$E_2' = k_e E_2 = k_e \frac{1}{k_e} E_1 = E_1$$

$$E_2' = k_e E_2 = k_e \frac{1}{k_e} E_1 = E_1$$

$$E_2' = k_e E_2 = k_e \frac{1}{k_e} E_1 = E_1$$

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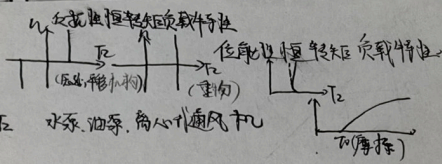
八. 生产机械负载类型 P_{22}

$T-T_L = J \frac{d\omega}{dt}$ (由关系确定)

恒转矩 $T_L = k_n$ (起重机)

通风机 $T_L = k_n \omega^2$

恒功率 $T_L = \frac{k}{\omega}$



九. 直流电动机

$n = \frac{U}{C_e \Phi} - \frac{R_a}{C_e \Phi} I_a$

$n = \frac{U}{C_e \Phi} - \frac{R_a}{C_e \Phi} I_a$

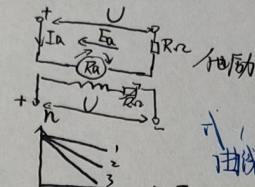
$T_e = C_T \Phi I_a$

$E_a = C_e \Phi n$

$U = E_a + I_a R_a$

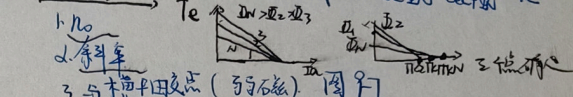
固有机械特性与人为机械特性 (改变电压电阻)

调速电阻, 降电压, 弱磁, P_{22} 分析题 图 9.4



由式 (9-8) 得

$$n = \frac{U}{C_e \Phi} - \frac{R_a}{C_e \Phi} I_a$$



$I_k = \frac{U}{R_a} (n=0)$ 堵转

$C_T \Phi I_k = T_k$ 堵转

十. Chap 10. 人为机械特性分析 (去年)

2. 人为机械特性 $P_{86.8}$

降低电压

转子串电阻 R_r

转子串电抗/电阻 X_r/R_r

转子接入并联阻抗

对于特殊点的分析

$n, n_s, T_{max}, S_m, T_{st}$

斜率为正

$K_T = \frac{T_{max}}{T_n}$

$K_s = \frac{S_m}{S_n}$

$T_e = \frac{m}{2s} \frac{U^2 R_r}{(R_s + R_r)^2 + (X_s + X_r)^2}$

$T_e = \frac{m}{2s} \frac{U^2 R_r}{(R_s + R_r)^2 + (X_s + X_r)^2}$

$T_e = \frac{m}{2s} \frac{U^2 R_r}{(R_s + R_r)^2 + (X_s + X_r)^2}$

3. 制动方式

回馈制动

能耗制动

反接制动

无级平滑制动

直接启动

减压启动

软启动

十一. 启动

直接启动

减压启动

软启动

回馈制动

能耗制动

反接制动

十二. 变频

变频调速

变频调速

变频调速

2. 稳态判断 P_{31}, P_{32}

必要: 机械特性曲线与负载特性曲线有交点

充分: 扰动 P_{32} 后从 B 回到 A

扰动 P_{32} 后从 B 回到 A

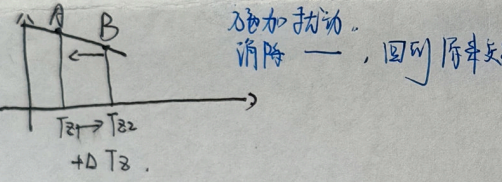


图 9.9

稳态的判断

线性/上翘

3. 制动方式

回馈制动

能耗制动

反接制动

4. 调速方式

调速范围

静差率

平滑性

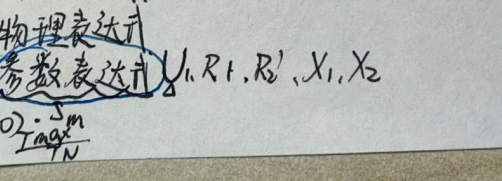
P_{31}

十. 异步机

1. 固有机械特性 $n = f(T_e)$

物理表达式

参数表达式 U, R_1, R_2', X_1, X_2



十二. 变频

变频调速

变频调速

变频调速

